INTERLOCKS

Raw Feed Transfer Pump A (UX-00201)

- Tank high high level switch LSHH-2002 healthy
- Discharge pressure PI-2008 > LoLo
- Pump discharge temperature TI-2007 < HiHi
- Discharge low low pressure switch PSLL-2010 healthy
- Transfer flow FQI-2009 > Lo

Raw Feed Transfer Pump B (UX-00202)

- Tank high high level switch LSHH-2002 healthy
- Discharge pressure PI-2008 > LoLo
- Pump discharge temperature TI-2007 < HiHi
- Discharge low low pressure switch PSLL-2010 healthy
- Transfer flow FQI-2009 > Lo

Primary Reactor Agitator (UX-00501)

- Reactor feed pressure PI-5021 > LoLo
- Reactor feed temperature TI-5022 < HiHi
- Reactor feed flow FI-5023 > Lo
- Reactor outlet pressure PI-5024 < HiHi
- Reactor high high temperature trip UZ-5025 healthy
- Jacket cooling water flow FI-5008 > Lo

INSTRUMENT LIST

Tag	Unit	Range	CtrlLo	CtrlHi	LoLo	Lo	Hi	HiHi
TI-2007	deg C	0-100	15	45	5	10	55	65
PI-2008	barg	0-10	2.0	6.0	1.5	2.5	6.5	8.0
FQI-2009	kg/h	0-10000	5000	8000	2000	3000	8500	9500
AI-3010	ppm	0-500	50	300	20	40	350	450
TI-3021	deg C	0-100	10	40	3	5	50	60
FI-4011	kg/h	0-10000	5000	8000	2000	3000	8500	9500
PI-4021	barg	0-10	2.0	6.0	1.0	1.5	6.5	8.0
FI-5008	m3/h	0-40	15	25	8	12	30	35

Product Cooler Fan (UX-00701)

- Cooler inlet pressure PI-7021 > LoLo
- Cooler inlet low low pressure trip UZ-7004 healthy
- Product flow FI-7006 > Lo
- Cooler outlet temperature TI-7005 < Hi

Product Loadout Pump A (UX-00702)

- Storage tank level LI-7003 > LoLo
- Tank high high level switch LSHH-7027 healthy
- Loadout discharge pressure PI-7023 > LoLo
- Loadout low low pressure trip UZ-7008 healthy
- Loadout flow FQI-7009 > Lo
- Product quality AI-7002 < Hi

Product Loadout Pump B (UX-00703)

- Storage tank level LI-7003 > LoLo
- Tank high high level switch LSHH-7027 healthy
- Loadout discharge pressure PI-7023 > LoLo
- Loadout low low pressure trip UZ-7008 healthy
- Loadout flow FQI-7009 > Lo
- Product quality AI-7002 < Hi

INSTRUMENT LIST

Tag	Unit	Range	CtrlLo	CtrlHi	LoLo	Lo	Hi	HiHi
PI-5021	barg	0-10	1.5	5.0	0.8	1.2	5.5	7.0
TI-5022	deg C	0-150	70	95	50	60	105	110
FI-5023	m3/h	0-20	6	14	2	4	16	18
AI-6008	ppm	0-1000	100	600	50	80	700	900
AI-7002	ppm	0-100	5	50	2	10	45	60
LI-7003	pct	0-100	20	80	10	15	85	92
TI-8021	deg C	0-100	20	40	10	15	45	55

Cooling Water Cooler Fan (UX-00801)

- Cooling water return temperature TI-8004 < HiHi
- Cooling water return pressure PI-8005 > LoLo
- Return low low flow switch FSLL-8006 healthy
- Cooler outlet temperature TI-8026 < HiHi

Cooling Water Circulation Pump A (UX-00802)

- Supply low low pressure switch PSLL-8011 healthy
- Pump discharge temperature TI-8021 < HiHi
- Flash receiver level LI-8003 > LoLo
- Cooling water supply temperature TI-8010 < HiHi

Cooling Water Circulation Pump B (UX-00803)

- Supply low low pressure switch PSLL-8011 healthy
- Pump discharge temperature TI-8021 < HiHi
- Flash receiver level LI-8003 > LoLo
- Cooling water supply temperature TI-8010 < HiHi

Tag No.	Equipment	Interlock Condition	Interlock Type		
			ST	P	SA
UX-00201	Raw Feed Transfer Pump A	Tank high high level switch LSHH-2002 healthy		X	
		Discharge pressure PI-2008 > LoLo		X	
		Pump discharge temperature TI-2007 < HiHi		X	
		Discharge low low pressure switch PSLL-2010 healthy	X		
		Transfer flow FQI-2009 > Lo		X	
UX-00202	Raw Feed Transfer Pump B	Tank high high level switch LSHH-2002 healthy		X	
		Discharge pressure PI-2008 > LoLo		X	
		Pump discharge temperature TI-2007 < HiHi		X	
		Discharge low low pressure switch PSLL-2010 healthy	X		
		Transfer flow FQI-2009 > Lo		X	
UX-00501	Primary Reactor Agitator	Reactor feed pressure PI-5021 > LoLo		X	
		Reactor feed temperature TI-5022 < HiHi		X	
		Reactor feed flow FI-5023 > Lo		X	
		Reactor outlet pressure PI-5024 < HiHi		X	
		Reactor high high temperature trip UZ-5025 healthy	X		
		Jacket cooling water flow FI-5008 > Lo		X	
UX-00701	Product Cooler Fan	Cooler inlet pressure PI-7021 > LoLo		X	
		Cooler inlet low low pressure trip UZ-7004 healthy	X		
		Product flow FI-7006 > Lo		X	
		Cooler outlet temperature TI-7005 < Hi			X
UX-00702	Product Loadout Pump A	Storage tank level LI-7003 > LoLo		X	
		Tank high high level switch LSHH-7027 healthy		X	
		Loadout discharge pressure PI-7023 > LoLo		X	
		Loadout low low pressure trip UZ-7008 healthy	X		
		Loadout flow FQI-7009 > Lo		X	
		Product quality AI-7002 < Hi			X
UX-00703	Product Loadout Pump B	Storage tank level LI-7003 > LoLo		X	
		Tank high high level switch LSHH-7027 healthy		X	
		Loadout discharge pressure PI-7023 > LoLo		X	
		Loadout low low pressure trip UZ-7008 healthy	X		
		Loadout flow FQI-7009 > Lo		X	
		Product quality AI-7002 < Hi			X
UX-00801	Cooling Water Cooler Fan	Cooling water return temperature TI-8004 < HiHi		X	
		Cooling water return pressure PI-8005 > LoLo		X	
		Return low low flow switch FSLL-8006 healthy	X		
		Cooler outlet temperature TI-8026 < HiHi		X	
UX-00802	Cooling Water Circulation Pump A	Supply low low pressure switch PSLL-8011 healthy	X		
		Pump discharge temperature TI-8021 < HiHi		X	
		Flash receiver level LI-8003 > LoLo		X	
		Cooling water supply temperature TI-8010 < HiHi		X	

Tag No.	Equipment	Interlock Condition	Interlock Type		
			ST	P	SA
UX-00803	Cooling Water Circulation Pump B	Supply low low pressure switch PSLL-8011 healthy	X		
		Pump discharge temperature TI-8021 < HiHi		X	
		Flash receiver level LI-8003 > LoLo		X	
		Cooling water supply temperature TI-8010 < HiHi		X	

The Cooling Water Circulation Pump A (P-8010A) operates under specific interlock conditions to ensure safe and reliable performance. The pump discharge temperature (TI-8021) must remain below the HiHi threshold, the flash receiver level (LI-8003) must remain above the LoLo threshold and the cooling water supply temperature (TI-8010) must remain below the HiHi threshold. Additionally, the supply low low pressure switch (PSLL-8011) must be healthy; a fall below that setting trips the drive. All of these permissives are held in interlock balloon UX-00802 and the starter cannot be energised until every one of them is satisfied.